

Chapter 5 – Regression Analysis

Detailed PPT for Sections 5.1, 5.2 and 5.3

Introduction to Regression • Correlation, Regression & Causation • Introduction to Linear Regression

Includes concepts, equations, numerical problems, step-by-step solutions and exam-oriented questions

Based on the uploaded S. Sridhar & M. Vijayalakshmi textbook, Chapter 5.

Chapter 5 – Learning Objectives

Understand regression as supervised learning for continuous/quantitative variables.

Distinguish correlation, regression and causation.

Identify linear and nonlinear relationships.

Understand different regression methods.

Explain limitations such as outliers, missing data and multicollinearity.

Construct and interpret a simple linear regression model.

Understand residuals and the OLS criterion.

Calculate regression coefficients and make predictions.

Represent linear regression using matrix notation.

5.1 Introduction to Regression

Regression models the relationship between one or more independent/predictor variables and a dependent/response variable.

Given training data $D = \{(x_i, y_i)\}$, regression learns a relationship that can be used for prediction.

Independent variable: explanatory variable, predictor, covariate or input.

Dependent variable: response, target or output.

Regression is primarily used to predict continuous or quantitative variables.

5.1 – Questions Addressed by Regression

What is the relationship between variables?

How strong is the relationship?

Is the relationship linear or nonlinear?

Which attributes are relevant?

How much does each relevant attribute contribute?

Can the learned relationship be used to predict a response for new input values?

5.1 – Regression Applications

Application	Response predicted
Sales prediction	Sales/revenue
Bond valuation	Bond value
Insurance	Premium/claim-related quantitative value
Agriculture	Crop yield
Real estate	Property price
General forecasting	Future quantitative measurement

Worked Problem 1 – Identify Regression Variables

Problem: A college wants to predict a student's final examination marks from attendance percentage.

Identify the independent and dependent variables.

State whether this is a regression problem.

Suggest another possible predictor.

Problem 1 – Solution

Independent/predictor variable x = attendance percentage.

Dependent/response variable y = final examination marks.

Yes. It is a regression problem because marks are quantitative/continuous.

Another predictor could be internal assessment marks, study hours or assignment score.

Example model: $\text{Marks} = f(\text{Attendance}, \text{Study Hours}, \text{Internal Marks})$.

Worked Problem 2 – Regression or Classification?

Determine whether each task is regression or classification:

1. Predict house price from area and location.
2. Predict whether an email is spam or not spam.
3. Predict crop yield in tonnes/hectare.
4. Predict whether a customer will churn: Yes/No.
5. Predict monthly electricity consumption in kWh.

Problem 2 – Solution

Task	Type	Reason
House price	Regression	Continuous price
Spam/Not spam	Classification	Categorical label
Crop yield	Regression	Quantitative value
Customer churn	Classification	Binary category
Electricity consumption	Regression	Continuous quantity

5.1 – General Regression Structure

Input/predictor variables $X \rightarrow$ Regression model $\rightarrow$ Response Y

The model estimates a mathematical relationship between predictors and response.

For a new predictor value, the learned relationship provides a predicted response.

With one predictor: $y = f(x)$.

With several predictors: $y = f(x_1, x_2, \dots, x_p)$.

5.2 Correlation, Regression and Causation

Correlation measures the strength and direction of association between variables.

Regression models/predicts one variable using one or more variables.

Causation asks whether a change in one variable actually produces a change in another.

Correlation or regression alone does not establish causation.

Scatter plots are useful for visually exploring relationships.

Correlation – Positive, Negative and Random

Positive relationship: as x increases, y tends to increase.

Negative relationship: as x increases, y tends to decrease.

Random/no clear relationship: points show no systematic trend.

The Pearson correlation coefficient r is emphasized in the chapter.

r indicates direction and strength of linear association.

Correlation Coefficient – Interpretation

r value / pattern	Interpretation
+1	Perfect positive linear association
Close to +1	Strong positive linear association
0	No linear association
Close to -1	Strong negative linear association
-1	Perfect negative linear association
Important	Correlation measures association; it does not prove causation.

Worked Problem 3 – Calculate Pearson Correlation

Given $x = [1, 2, 3, 4, 5]$

and $y = [2, 4, 5, 4, 5]$.

Use:

$$r = [n\sum xy - \sum x \sum y] / \sqrt{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]}$$

Calculate r and interpret the result.

Problem 3 – Step 1: Calculate Required Sums

$$n = 5$$

$$\Sigma x = 15$$

$$\Sigma y = 20$$

$$\Sigma x^2 = 55$$

$$\Sigma y^2 = 4+16+25+16+25 = 86$$

$$\Sigma xy = 2+8+15+16+25 = 66$$

Problem 3 – Step 2: Calculate r

$$\text{Numerator} = 5(66) - (15)(20) = 330 - 300 = 30.$$

$$\text{First denominator term} = 5(55) - 15^2 = 275 - 225 = 50.$$

$$\text{Second denominator term} = 5(86) - 20^2 = 430 - 400 = 30.$$

$$\text{Denominator} = \sqrt{(50 \times 30)} = \sqrt{1500} \approx 38.73.$$

$$r = 30/38.73 \approx 0.775$$

Interpretation: a moderately strong positive linear association exists.

Regression vs Correlation

Correlation is symmetric: $\text{correlation}(x,y) = \text{correlation}(y,x)$.

Regression is directional: one variable is modeled/predicted from another.

Correlation summarizes association; regression provides an equation for prediction.

High correlation can support a useful linear model, but it does not by itself prove causation.

Regression coefficients depend on the chosen response and predictor variables.

Regression vs Causation – Example

Suppose ice-cream sales and the number of people at a beach are positively correlated.

One cannot conclude that ice-cream sales cause people to visit the beach.

A third factor such as hot weather can influence both variables.

Therefore, observed regression/correlation may reflect common or omitted factors.

Domain knowledge and appropriate experimental/study design are required for causal claims.

Linearity and Non-Linearity

Linear relationship: $y = ax + b$.

Nonlinear relationships cannot always be represented adequately by a straight line.

Examples given in the chapter include exponential relationships such as $y = ae^{(bx)}$.

Power relationships are another nonlinear form.

The regression method should match the structure of the relationship.

Worked Problem 4 – Identify the Relationship

For each model, identify whether the relationship is linear or nonlinear:

1. $y = 4x + 7$

2. $y = 2e^{(0.5x)}$

3. $y = 3x^2 + 2x + 1$

4. $y = 5x + 2$

Problem 4 – Solution

Model	Type	Reason
$y=4x+7$	Linear	x has first-degree coefficient
$y=2e^{(0.5x)}$	Nonlinear	Exponential function
$y=3x^2+2x+1$	Nonlinear relationship	Contains x^2 term
$y=5x+2$	Linear	Straight-line form

Types of Regression Methods

The chapter classifies regression methods into linear, nonlinear and logistic regression.

Linear regression: one or more predictors with a linear relationship.

Simple linear regression: one predictor.

Multiple linear regression: two or more independent variables and one dependent variable.

Polynomial regression: polynomial terms represent a curved relationship.

Logistic regression: categorical/binary response; used as a regression-based classifier.

Ridge and LASSO are regularized regression variants discussed later in the chapter.

Regression Methods – Comparison

Method	Predictors / form	Typical purpose
Simple linear	One predictor; $y = a_0 + a_1x$	Continuous prediction
Multiple linear	Several predictors	Continuous prediction using multiple factors
Polynomial	Polynomial terms	Curved/nonlinear relationship
Logistic	Logit/sigmoid form	Binary/categorical response
Ridge/LASSO	Penalized coefficients	Regularized regression

5.2 – Limitations and Issues in Regression

Outliers can strongly influence the fitted regression line.

Missing data may make a model unsuitable for the sampled dataset.

Multicollinearity occurs when explanatory variables are highly correlated with one another.

Multicollinearity can bias/unstabilize regression coefficients.

The chapter mentions removing problematic variables and using tolerance as part of addressing this issue.

A rule of thumb stated in the text is at least about 20 cases per explanatory variable, with fewer only in extreme cases.

Worked Problem 5 – Detect a Potential Outlier

Suppose a fitted model predicts salaries:

$$\hat{y} = 20 + 3x$$

For $x = 5$, predicted salary = 35.

Observed salary is 80.

Find the residual and explain why the observation may need investigation.

Problem 5 – Solution

Predicted salary: $\hat{y} = 20 + 3(5) = 35$.

Residual = $y - \hat{y} = 80 - 35 = +45$.

The residual is very large compared with the prediction.

Such an observation may be an outlier or may indicate an omitted explanatory factor.

Outliers should be investigated rather than automatically deleted.

5.3 Introduction to Linear Regression

Simple linear regression fits a straight line through scattered data points.

Model: $y = a_0 + a_1x + \varepsilon$.

a_0 = intercept; a_1 = slope; ε = prediction error.

Predicted response: $\hat{y} = a_0 + a_1x$.

The objective is to obtain coefficients giving the best fit under the selected criterion.

Components of the Linear Regression Model

Symbol	Meaning	Interpretation
x	Predictor	Input variable
y	Observed response	Actual output
a_0	Intercept	Predicted y at $x=0$, if meaningful
a_1	Slope	Change in predicted y for one-unit x increase
ϵ	Error	Random deviation
$\hat{y}$	Predicted response	Fitted model output

Assumptions of Linear Regression

Observations are treated as random and mutually independent.

The relationship between predictor and response is represented by a linear model.

Errors are treated as mutually independent with a common distribution.

The chapter discusses normal errors with zero mean and constant variance.

Unknown regression parameters are treated as constants.

These assumptions should be examined when applying the model to real data.

Residuals

Residual: $e_i = y_i - \hat{y}_i$.

Positive residual means the observed value is above the fitted line.

Negative residual means the observed value is below the fitted line.

Zero residual means the prediction exactly equals the observation.

Residuals are the basis for evaluating the fit and constructing the OLS objective.

Why Squared Errors?

Raw errors can cancel: $+4 - 3 + 2 - 3 = 0$.

A zero sum does not imply perfect prediction.

Absolute errors avoid cancellation but do not penalize large errors as strongly.

Squared errors are always non-negative and give greater weight to large deviations.

OLS therefore minimizes the sum of squared residuals.

Ordinary Least Squares (OLS)

OLS chooses a_0 and a_1 to minimize:

$$J(a_0, a_1) = \sum [y_i - (a_0 + a_1 x_i)]^2$$

The fitted line is called the line of best fit under the OLS criterion.

Differentiate the objective with respect to a_0 and a_1 .

Set both derivatives to zero and solve the normal equations.

Simple Linear Regression – Formulae

Slope: $a_1 = [\sum x_i y_i - n \bar{x} \bar{y}] / [\sum x_i^2 - n \bar{x}^2]$

Equivalent centered form: $a_1 = \sum (x_i - \bar{x})(y_i - \bar{y}) / \sum (x_i - \bar{x})^2$

Intercept: $a_0 = \bar{y} - a_1 \bar{x}$

Prediction: $\hat{y} = a_0 + a_1 x$

Worked Problem 6 – Complete Textbook-Style Calculation

Given $x = 1, 2, 3, 4, 5$ and $y = 1.2, 1.8, 2.6, 3.2, 3.8$.

Find the regression equation.

Then predict y for $x = 7$ and $x = 12$.

Problem 6 – Calculation Table

x	y	x ²	xy
1	1.2	1	1.2
2	1.8	4	3.6
3	2.6	9	7.8
4	3.2	16	12.8
5	3.8	25	19.0
Σx=15	Σy=12.6	Σx ² =55	Σxy=44.4

Problem 6 – Solution: Slope and Intercept

$$n = 5, \bar{x} = 15/5 = 3, \bar{y} = 12.6/5 = 2.52.$$

$$\begin{aligned} a_1 &= [44.4 - 5(3)(2.52)]/[55 - 5(3^2)] \\ &= 6.6/10 = 0.66 \end{aligned}$$

$$a_0 = \bar{y} - a_1\bar{x} = 2.52 - (0.66)(3) = 0.54.$$

$$\text{Regression equation: } \hat{y} = 0.54 + 0.66x$$

Problem 6 – Prediction

For $x = 7$:

$$\hat{y} = 0.54 + 0.66(7) = 5.16$$

For $x = 12$:

$$\hat{y} = 0.54 + 0.66(12) = 8.46$$

Interpretation: slope 0.66 means predicted y increases by 0.66 for every one-unit increase in x .

Worked Problem 7 – Compare Two Candidate Lines

Data: (1,3), (2,5), (3,7).

Line A: $\hat{y} = 1 + 2x \rightarrow$ predictions 3,5,7 $\rightarrow$ SSE = 0.

Line B: $\hat{y} = 2 + 1.5x \rightarrow$ predictions 3.5,5,6.5 $\rightarrow$ SSE = 0.50.

Since $SSE(A) < SSE(B)$, Line A is better under OLS.

Worked Problem 8 – Residual, SSE, MSE and RMSE

Actual $y = [10, 12, 15, 18]$

Predicted $\hat{y} = [11, 11, 14, 20]$

Residuals = $[-1, +1, +1, -2]$

$SSE = 1 + 1 + 1 + 4 = 7$

$MSE = 7/4 = 1.75$

$RMSE = \sqrt{1.75} \approx 1.323$

Linear Regression in Matrix Form

$$Y = X\alpha + \varepsilon$$

Y = response vector.

X = design matrix with a column of 1s for the intercept.

α = coefficient vector $[a_0, a_1]^T$.

ε = error vector.

OLS estimate: $\hat{\alpha} = (X^T X)^{-1} X^T Y$, when the inverse exists.

Worked Problem 9 – Matrix Solution

For $x = 1, 2, 3$ and $y = 2, 4, 5$:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$Y = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 11 \\ 25 \end{bmatrix}$$

$$(X^T X)^{-1} = (1/6) \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$$

$$\hat{\alpha} = \begin{bmatrix} 0.6667 \\ 1.5 \end{bmatrix}$$

Therefore $\hat{y} \approx 0.667 + 1.5x$.

Integrated Problem – From Data to Prediction

Given $x = [2, 4, 6, 8]$ and $y = [5, 9, 12, 17]$.

Given $\bar{x}=5$, $\bar{y}=10.75$, $\Sigma(x-\bar{x})(y-\bar{y})=39$ and $\Sigma(x-\bar{x})^2=20$.

$$a_1 = 39/20 = 1.95$$

$$a_0 = 10.75 - 1.95(5) = 1.00$$

Regression equation: $\hat{y} = 1 + 1.95x$

For $x=10$, $\hat{y}=20.5$

5.1–5.3 Integrated Concept Map

Regression → predicts quantitative response.

5.1: What is regression? Applications, predictors and response.

5.2: How are correlation, regression and causation different?

5.2: Linear vs nonlinear relationships; types and limitations.

5.3: Simple linear regression, residuals, OLS, coefficients and prediction.

5.3: Matrix formulation provides a compact general representation.

Important Formula Sheet – 5.1 to 5.3

Regression: $y = f(x)$.

Linear model: $y = a_0 + a_1x + \varepsilon$.

Prediction: $\hat{y} = a_0 + a_1x$.

Residual: $e_i = y_i - \hat{y}_i$.

OLS: $J = \sum [y_i - (a_0 + a_1x_i)]^2$.

Slope: $a_1 = [\sum xy - n \bar{x}\bar{y}] / [\sum x^2 - n \bar{x}^2]$.

Intercept: $a_0 = \bar{y} - a_1\bar{x}$.

Pearson $r = [n\sum xy - \sum x \sum y] / \sqrt{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]}$.

Matrix OLS: $\hat{\alpha} = (X^T X)^{-1} X^T Y$.

Common Student Mistakes

Calling every prediction problem regression even when the target is categorical.

Confusing correlation with regression.

Assuming correlation/regression proves causation.

Using a linear model when the relationship is clearly nonlinear.

Confusing observed y with predicted $\hat{y}$.

Using raw error sum instead of squared error for OLS.

Misinterpreting the intercept when $x=0$ has no practical meaning.

Ignoring outliers, missing values or multicollinearity.

Exam-Oriented Questions – L2/L3

1. Define regression and distinguish predictor and response variables.
2. Explain positive, negative and zero correlation with examples.
3. Differentiate correlation, regression and causation.
4. Explain simple and multiple linear regression.
5. Explain residual and OLS.
6. Calculate Pearson correlation for a small dataset.
7. Calculate slope and intercept for a given dataset.
8. Predict the response for a specified x .

Exam-Oriented Questions – L4

1. A dataset contains study hours and marks. Construct a regression model and interpret the slope.
2. Compare two candidate regression lines using SSE.
3. Given actual and predicted values, calculate residuals, SSE, MSE and RMSE and comment on model fit.
4. A dataset contains highly correlated predictors. Explain the problem and suggest an appropriate diagnostic/strategy.
5. Given a nonlinear scatter pattern, justify whether linear regression is appropriate.
6. Use matrix notation to derive/compute the OLS coefficient vector.

Final Revision – 5.1, 5.2 and 5.3

5.1 → Regression predicts continuous/quantitative responses.

5.2 → Correlation measures association; regression supports prediction; causation requires stronger evidence.

5.2 → Choose linear/nonlinear methods according to the relationship.

5.2 → Watch for outliers, missing data and multicollinearity.

5.3 → Simple linear regression: $y = a_0 + a_1x + \epsilon$.

5.3 → Residual = observed – predicted.

5.3 → OLS minimizes squared residuals.

5.3 → Calculate coefficients, predictions and errors systematically.

5.3 → Matrix form generalizes the computation.